BIG DATA TECHNOLOGY LAB (BDTL) · PRACTICAL REPORT

Bucket 8 · Apache Pig Experiments 2 and 3

Submitted by

Adesh Patankar

Experiment	Title and learning objective
Experiment 2	**Group and Aggregate Data** Use GROUP BY on department or subject and calculate average marks or total students.
Experiment 3	**Sort Data** Use ORDER BY or LIMIT to sort results by marks or names. Compare ascending vs. descending order.

Platform: Apache Pig 0.18.0 on Apache Hadoop 3.3.6 (pseudo-distributed HDFS + YARN), Ubuntu 24.04 LTS. All execution evidence in this report consists of screenshots of the actual terminal session in which the experiments were run.

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Bucket 8 – Experiment 2: Group and Aggregate Data

A. Experiment Details

Field	Details
Bucket	8
Sub-experiment No.	2
Title	Group and Aggregate Data
Description / Learning Objective (bucket list)	Use GROUP BY on department or subject and calculate average marks or total students.
Student	Adesh Patankar
Subject	Big Data Technology Lab (BDTL)
Tool / Technology	Apache Pig 0.18.0 (Pig Latin) on Apache Hadoop 3.3.6

B. Aim

To group records with `GROUP BY` and compute aggregates: the total number of students in each department, the average (and maximum/minimum) marks in each subject, and the average marks of each department.

C. Theory

C.1 Apache Pig

Apache Pig is a high-level platform on Hadoop for analysing large datasets as data flows written in **Pig Latin**. The Pig engine parses a script into a logical plan, optimises it, compiles it into MapReduce jobs and runs them on YARN, reading and writing HDFS. A **relation** is a bag of **tuples**; each tuple has typed **fields**. `LOAD ... USING PigStorage(',') AS (schema)` reads a comma-separated file with named, typed columns. Statements are evaluated lazily – jobs run only when `DUMP` or `STORE` needs a result. Pig runs here in **MapReduce mode** (`pig -x mapreduce`).

C.2 GROUP and aggregate functions

`g = GROUP r BY key;` produces one tuple per distinct key: `(group, {bag of all tuples of r with that key})`. The key field is always named `group` and the bag carries the name of the input relation. Aggregates are then applied per group with `FOREACH g GENERATE group, FUNC(r.field)`.

Function	Returns
`COUNT(bag)`	number of tuples (long)
`AVG(bag.field)`	average (double)
`MAX` / `MIN`	largest / smallest value
`SUM`	total

These functions are *algebraic*, so Pig adds a **combiner** that pre-aggregates on the map side (feature `GROUP_BY,COMBINER`). To aggregate marks by department, the two datasets are first joined

and then grouped on `students::dept`.

D. Requirements

Item	Used in this experiment
Host machine	Apple MacBook Air (Apple Silicon, arm64) running macOS
Virtual machine	Multipass VM `bigdata` – Ubuntu 24.04 LTS (arm64), 2 vCPU, 4 GB RAM, 2 GB swap
Java	OpenJDK 8
Hadoop	Apache Hadoop 3.3.6, pseudo-distributed: HDFS (`hdfs://localhost:9000`), YARN and the MapReduce JobHistory server
Pig	Apache Pig 0.18.0 (Hadoop 3 build), run in **MapReduce mode**: `pig -x mapreduce -f <script>.pig`; `PIG_HOME`, `PATH` and `PIG_CLASSPATH=$HADOOP_HOME/etc/hadoop` set in `~/.bashrc`
Logging	A custom `log4j.properties` (passed with `-4`) keeps Pig's job progress, job statistics and errors on the console and hides routine INFO messages
Datasets	`students.txt` (8 rows) and `marks.txt` (17 rows) in `/bdt/bucket8/`
Runner	`bdt/bucket8/run_bucket8.sh` – prints each Pig script (`cat`) and runs it with `pig -x mapreduce -4 ~/bdt/pig/log4j.properties -f <script>`

E. Procedure and Implementation

Step 1 – Datasets and upload to HDFS

Both datasets were created for this experiment. `students.txt` has 8 students (id, name, department). `marks.txt` has 17 rows (id, subject, marks) for two subjects; it deliberately contains two exact duplicate rows (`103,DBMS,91` and `106,BigData,74`), one row for an id with no student (`109`), and student 108 (Isha) has no marks.

bucket8/data/students.txt

```
101,Aarav,Computer
102,Diya,IT
103,Kabir,Computer
104,Meera,Mechanical
105,Rohan,IT
106,Sneha,Computer
107,Arjun,Mechanical
108,Isha,IT
```

bucket8/data/marks.txt

```
101,DBMS,78
101,BigData,85
102,DBMS,45
102,BigData,62
103,DBMS,91
103,BigData,88
104,DBMS,38
104,BigData,55
105,DBMS,67
105,BigData,49
106,DBMS,82
106,BigData,74
```

```
107,DBMS,59
107,BigData,41
103,DBMS,91
106,BigData,74
109,DBMS,70
```

Shell commands

```
pig -version
jps
cat data/students.txt
cat data/marks.txt
hdfs dfs -mkdir -p /bdt/bucket8
hdfs dfs -put -f data/students.txt data/marks.txt /bdt/bucket8/
hdfs dfs -ls /bdt/bucket8
```

Step 2 – Pig Latin script

bucket8/pig/02_group.pig

```
-- Bucket 8 | Sub-Experiment 2 : Group and aggregate data
students = LOAD '/bdt/bucket8/students.txt' USING PigStorage(',') AS (id:int, name:chararray,
dept:chararray);
marks    = LOAD '/bdt/bucket8/marks.txt'    USING PigStorage(',') AS (id:int, subject:chararray,
marks:int);

-- total students in each department
by_dept  = GROUP students BY dept;
dept_cnt = FOREACH by_dept GENERATE group AS dept, COUNT(students) AS total_students;
DESCRIBE by_dept;
DUMP dept_cnt;

-- average, maximum and minimum marks in each subject
by_subj  = GROUP marks BY subject;
subj_avg = FOREACH by_subj GENERATE group AS subject, COUNT(marks) AS entries,
           AVG(marks.marks) AS avg_marks, MAX(marks.marks) AS max_marks, MIN(marks.marks) AS
min_marks;
DUMP subj_avg;

-- average marks of each department (join first, then group)
joined   = JOIN students BY id, marks BY id;
by_dept2 = GROUP joined BY students::dept;
dept_avg = FOREACH by_dept2 GENERATE group AS dept, AVG(joined.marks::marks) AS avg_marks;
DUMP dept_avg;
```

Command

```
pig -x mapreduce -4 ~/bdt/pig/log4j.properties -f pig/02_group.pig
```

F. Execution Screenshots

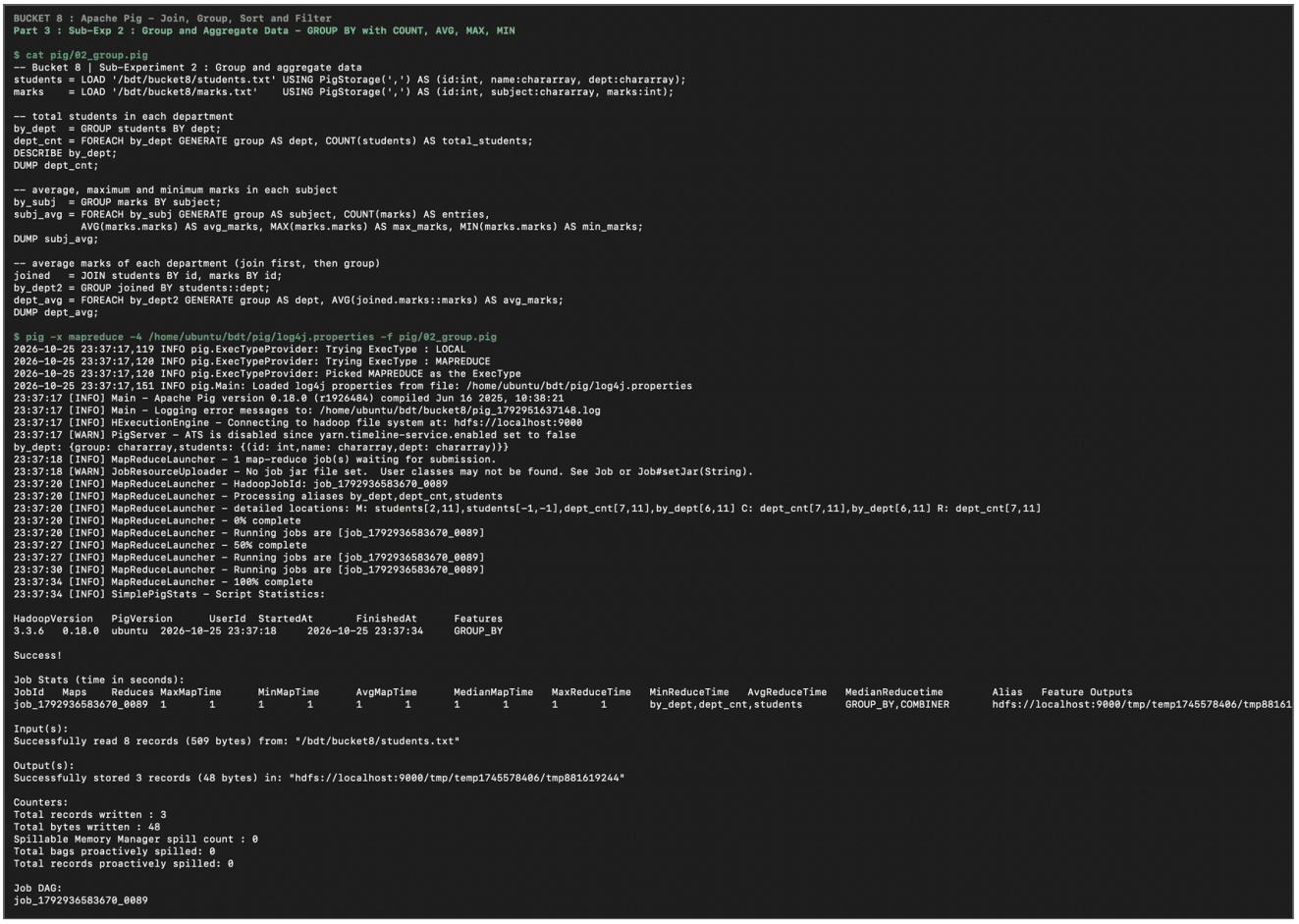
Screenshots of the Ubuntu VM terminal (`ubuntu@bigdata`) while `run_bucket8.sh` was executing. Only empty margins were trimmed.

```
BUCKET 8 : Apache Pig - Join, Group, Sort and Filter
Part 1 : Setup: environment check, sample datasets and upload to HDFS

$ pig -version
Apache Pig version 0.18.0 (r1926484)
compiled Jun 16 2025, 10:38:21
$ jps
1696 NameNode
52177 Jps
3377 JobHistoryServer
2483 ResourceManager
1916 DataNode
2157 SecondaryNameNode
2687 NodeManager
$ cat data/students.txt
101,Aarav,Computer
102,Diya,IT
103,Kabir,Computer
104,Meera,Mechanical
105,Rohan,IT
106,Sneha,Computer
107,Arjun,Mechanical
108,Isha,IT
$ cat data/marks.txt
101,DBMS,78
101,BigData,85
102,DBMS,45
102,BigData,62
103,DBMS,91
103,BigData,88
104,DBMS,38
104,BigData,55
105,DBMS,67
105,BigData,49
106,DBMS,82
106,BigData,74
107,DBMS,59
107,BigData,41
103,DBMS,91
106,BigData,74
109,DBMS,70
$ hdfs dfs -mkdir -p /bdt/bucket8
$ hdfs dfs -put -f data/students.txt data/marks.txt /bdt/bucket8/
$ hdfs dfs -ls /bdt/bucket8
Found 2 items
-rw-r--r--   1 ubuntu supergroup        228 2026-10-25 23:36 /bdt/bucket8/marks.txt
-rw-r--r--   1 ubuntu supergroup        136 2026-10-25 23:36 /bdt/bucket8/students.txt
```

Figure 2.1: Environment check, datasets and upload to HDFS

`pig -version` reports Apache Pig 0.18.0 and `jps` lists all Hadoop daemons including the JobHistoryServer. Both datasets are printed and copied to `/bdt/bucket8/` (marks.txt 228 B, students.txt 136 B).



```
BUCKET 8 : Apache Pig – Join, Group, Sort and Filter
Part 3 : Sub-Exp 2 : Group and Aggregate Data – GROUP BY with COUNT, AVG, MAX, MIN

$ cat pig/02_group.pig
-- Bucket 8 | Sub-Experiment 2 : Group and aggregate data
students = LOAD '/bdt/bucket8/students.txt' USING PigStorage(',') AS (id:int, name:chararray, dept:chararray);
marks    = LOAD '/bdt/bucket8/marks.txt'    USING PigStorage(',') AS (id:int, subject:chararray, marks:int);

-- total students in each department
by_dept  = GROUP students BY dept;
dept_cnt = FOREACH by_dept GENERATE group AS dept, COUNT(students) AS total_students;
DESCRIBE by_dept;
DUMP dept_cnt;

-- average, maximum and minimum marks in each subject
by_subj  = GROUP marks BY subject;
subj_avg = FOREACH by_subj GENERATE group AS subject, COUNT(marks) AS entries,
           AVG(marks.marks) AS avg_marks, MAX(marks.marks) AS max_marks, MIN(marks.marks) AS min_marks;
DUMP subj_avg;

-- average marks of each department (join first, then group)
joined   = JOIN students BY id, marks BY id;
by_dept2 = GROUP joined BY students::dept;
dept_avg = FOREACH by_dept2 GENERATE group AS dept, AVG(joined.marks::marks) AS avg_marks;
DUMP dept_avg;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/02_group.pig
2026-10-25 23:37:17,119 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 23:37:17,120 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 23:37:17,120 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 23:37:17,151 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
23:37:17 [INFO] Main – Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
23:37:17 [INFO] Main – Logging error messages to: /home/ubuntu/bdt/bucket8/pig_1792951637148.log
23:37:17 [INFO] HExecutionEngine – Connecting to hadoop file system at: hdfs://localhost:9000
23:37:17 [WARN] PigServer – ATS is disabled since yarn.timeline-service.enabled set to false
by_dept: {group: chararray,students: {(id: int,name: chararray,dept: chararray)}}
23:37:18 [INFO] MapReduceLauncher – 1 map-reduce job(s) waiting for submission.
23:37:18 [WARN] JobResourceUploader – No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:37:20 [INFO] MapReduceLauncher – HadoopJobId: job_1792936583670_0089
23:37:20 [INFO] MapReduceLauncher – Processing aliases by_dept,dept_cnt,students
23:37:20 [INFO] MapReduceLauncher – detailed locations: M: students[2,11],students[-1,-1],dept_cnt[7,11],by_dept[6,11] C: dept_cnt[7,11],by_dept[6,11] R: dept_cnt[7,11]
23:37:20 [INFO] MapReduceLauncher – 0% complete
23:37:20 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0089]
23:37:27 [INFO] MapReduceLauncher – 50% complete
23:37:27 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0089]
23:37:30 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0089]
23:37:34 [INFO] MapReduceLauncher – 100% complete
23:37:34 [INFO] SimplePigStats – Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 23:37:18     2026-10-25 23:37:34     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0089  1       1       1       1       1       1       1       1       by_dept,dept_cnt,students       GROUP_BY,COMBINER       hdfs://localhost:9000/tmp/temp1745578406/tmp88161

Input(s):
Successfully read 8 records (509 bytes) from: "/bdt/bucket8/students.txt"

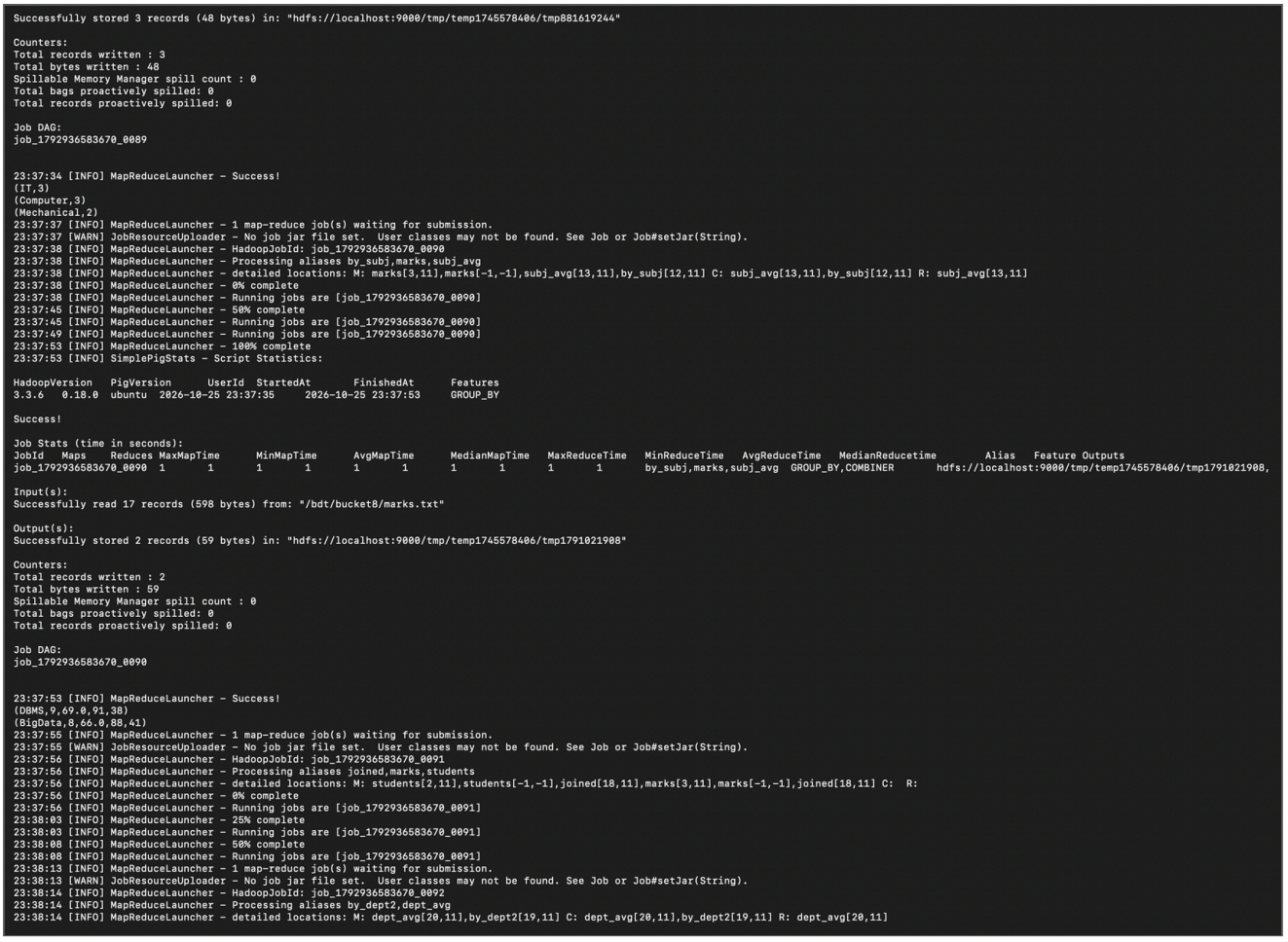
Output(s):
Successfully stored 3 records (48 bytes) in: "hdfs://localhost:9000/tmp/temp1745578406/tmp881619244"

Counters:
Total records written : 3
Total bytes written : 48
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0089
```

Figure 2.2: GROUP script, schema and the department-count job

The script is printed and run. `DESCRIBE by_dept` shows `{group: chararray, students: {(id, name, dept)}}`. Job `_0089` (GROUP_BY, COMBINER) reads 8 students and stores 3 records.



```
Successfully stored 3 records (48 bytes) in: "hdfs://localhost:9000/tmp/temp1745578406/tmp881619244"

Counters:
Total records written : 3
Total bytes written : 48
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0089


23:37:34 [INFO] MapReduceLauncher - Success!
(IT,3)
(Computer,3)
(Mechanical,2)
23:37:37 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:37:37 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:37:38 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0090
23:37:38 [INFO] MapReduceLauncher - Processing aliases by_subj,marks,subj_avg
23:37:38 [INFO] MapReduceLauncher - detailed locations: M: marks[3,11],marks[-1,-1],subj_avg[13,11],by_subj[12,11] C: subj_avg[13,11],by_subj[12,11] R: subj_avg[13,11]
23:37:38 [INFO] MapReduceLauncher - 0% complete
23:37:38 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0090]
23:37:45 [INFO] MapReduceLauncher - 50% complete
23:37:45 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0090]
23:37:49 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0090]
23:37:53 [INFO] MapReduceLauncher - 100% complete
23:37:53 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 23:37:35     2026-10-25 23:37:53     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0090  1       1       1       1       1       1       1       1       1       by_subj,marks,subj_avg  GROUP_BY,COMBINER       hdfs://localhost:9000/tmp/temp1745578406/tmp1791021908,

Input(s):
Successfully read 17 records (598 bytes) from: "/bdt/bucket8/marks.txt"

Output(s):
Successfully stored 2 records (59 bytes) in: "hdfs://localhost:9000/tmp/temp1745578406/tmp1791021908"

Counters:
Total records written : 2
Total bytes written : 59
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0090


23:37:53 [INFO] MapReduceLauncher - Success!
(DBMS,9,69.0,91,38)
(BigData,8,66.0,88,41)
23:37:55 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:37:55 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:37:56 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0091
23:37:56 [INFO] MapReduceLauncher - Processing aliases joined,marks,students
23:37:56 [INFO] MapReduceLauncher - detailed locations: M: students[2,11],students[-1,-1],joined[18,11],marks[3,11],marks[-1,-1],joined[18,11] C:  R:
23:37:56 [INFO] MapReduceLauncher - 0% complete
23:37:56 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0091]
23:38:03 [INFO] MapReduceLauncher - 25% complete
23:38:03 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0091]
23:38:08 [INFO] MapReduceLauncher - 50% complete
23:38:08 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0091]
23:38:13 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:38:13 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:38:14 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0092
23:38:14 [INFO] MapReduceLauncher - Processing aliases by_dept2,dept_avg
23:38:14 [INFO] MapReduceLauncher - detailed locations: M: dept_avg[20,11],by_dept2[19,11] C: dept_avg[20,11],by_dept2[19,11] R: dept_avg[20,11]
```

Figure 2.3: Total students per department and marks statistics per subject

Result `(IT,3) (Computer,3) (Mechanical,2)`. Job `_0090` groups the 17 marks rows by subject: `(DBMS,9,69.0,91,38)` and `(BigData,8,66.0,88,41)` (entries, average, maximum, minimum). The join for the department average starts.



Figure 2.4: Average marks per department

Two chained jobs (`_0091` HASH_JOIN → `_0092` GROUP_BY) produce `(IT,55.75)` `(Computer,82.875)` `(Mechanical,48.25)`. The script completes in 1 min 13 s.

G. Output and Results

Department	Total students	Average marks
Computer	3	82.875
IT	3	55.75
Mechanical	2	48.25

Subject	Entries	Average	Max	Min
DBMS	9	69.0	91	38
BigData	8	66.0	88	41

Check: IT = (45+62+67+49)/4 = 55.75; Mechanical = (38+55+59+41)/4 = 48.25; DBMS = 621/9 = 69.0. The averages include the duplicate rows in `marks.txt` (e.g. Computer = 663/8 = 82.875 counts Kabir's DBMS 91 twice), and the subject counts include the row for id 109, which has no student.

H. Result

> The objective was achieved. `GROUP BY` with `COUNT` gave the total students per department (Computer 3, IT 3, Mechanical 2), `GROUP BY subject` with `AVG/MAX/MIN` gave the subject statistics, and joining then grouping by department gave the average marks per department. All values were verified by hand.

I. Observations and Troubleshooting

Observations

- Each GROUP ran as one job with a combiner; the department average needed two jobs (join, then group).
- Duplicate input rows change aggregates: removing them with DISTINCT first would give Computer an average of 82.0 instead of 82.875.
- AVG returns a double (`69.0`) even when the inputs are integers.

Setup notes

- Apache Pig 0.18.0 (Hadoop 3 build) with `PIG_HOME`, `PATH` and `PIG_CLASSPATH` set in `~/.bashrc`; the MapReduce JobHistory server must be running because Pig requests job statistics from it.
- The warnings *No job jar file set*, *ATS is disabled* and *SchemaTupleBackend has already been initialized* are harmless in this pseudo-distributed setup.
- The script ran once without errors; no troubleshooting was required.

Bucket 8 – Experiment 3: Sort Data

A. Experiment Details

Field	Details
Bucket	8
Sub-experiment No.	3
Title	Sort Data
Description / Learning Objective (bucket list)	Use ORDER BY or LIMIT to sort results by marks or names. Compare ascending vs. descending order.
Student	Adesh Patankar
Subject	Big Data Technology Lab (BDTL)
Tool / Technology	Apache Pig 0.18.0 (Pig Latin) on Apache Hadoop 3.3.6

B. Aim

To sort student marks with `ORDER BY` in ascending and descending order, select the top results with `LIMIT`, and sort students alphabetically by name.

C. Theory

C.1 Apache Pig

Apache Pig is a high-level platform on Hadoop for analysing large datasets as data flows written in **Pig Latin**. The Pig engine parses a script into a logical plan, optimises it, compiles it into MapReduce jobs and runs them on YARN, reading and writing HDFS. A **relation** is a bag of **tuples**; each tuple has typed **fields**. `LOAD ... USING PigStorage(',') AS (schema)` reads a comma-separated file with named, typed columns. Statements are evaluated lazily – jobs run only when `DUMP` or `STORE` needs a result. Pig runs here in **MapReduce mode** (`pig -x mapreduce`).

C.2 ORDER BY and LIMIT

Statement	Meaning
`ORDER r BY f ASC;`	Sort from smallest to largest (default).
`ORDER r BY f DESC;`	Sort from largest to smallest.
`ORDER r BY f1 DESC, f2 ASC;`	Multi-key sort.
`LIMIT r n;`	Keep n tuples; after an ORDER BY it returns the top-n.

Pig implements a total-order sort in MapReduce with a **sampling job** that estimates the key distribution (feature `SAMPLER`) followed by an **ORDER_BY** job whose partitioner sends key ranges to reducers, so the output is globally sorted. Chararray fields sort lexicographically; ties keep no guaranteed order.

D. Requirements

Item	Used in this experiment
Host machine	Apple MacBook Air (Apple Silicon, arm64) running macOS
Virtual machine	Multipass VM `bigdata` – Ubuntu 24.04 LTS (arm64), 2 vCPU, 4 GB RAM, 2 GB swap
Java	OpenJDK 8
Hadoop	Apache Hadoop 3.3.6, pseudo-distributed: HDFS (`hdfs://localhost:9000`), YARN and the MapReduce JobHistory server
Pig	Apache Pig 0.18.0 (Hadoop 3 build), run in **MapReduce mode**: `pig -x mapreduce -f <script>.pig`; `PIG_HOME`, `PATH` and `PIG_CLASSPATH=$HADOOP_HOME/etc/hadoop` set in `~/.bashrc`
Logging	A custom `log4j.properties` (passed with `-4`) keeps Pig's job progress, job statistics and errors on the console and hides routine INFO messages
Datasets	`students.txt` (8 rows) and `marks.txt` (17 rows) in `/bdt/bucket8/`
Runner	`bdt/bucket8/run_bucket8.sh` – prints each Pig script (`cat`) and runs it with `pig -x mapreduce -4 ~/bdt/pig/log4j.properties -f <script>`

E. Procedure and Implementation

Step 1 – Datasets and upload to HDFS

Both datasets were created for this experiment. `students.txt` has 8 students (id, name, department). `marks.txt` has 17 rows (id, subject, marks) for two subjects; it deliberately contains two exact duplicate rows (`103,DBMS,91` and `106,BigData,74`), one row for an id with no student (`109`), and student 108 (Isha) has no marks.

bucket8/data/students.txt

```
101,Aarav,Computer
102,Diya,IT
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105,Rohan,IT
106,Sneha,Computer
107,Arjun,Mechanical
108,Isha,IT
```

bucket8/data/marks.txt

```
101,DBMS,78
101,BigData,85
102,DBMS,45
102,BigData,62
103,DBMS,91
103,BigData,88
104,DBMS,38
104,BigData,55
105,DBMS,67
105,BigData,49
106,DBMS,82
106,BigData,74
107,DBMS,59
107,BigData,41
103,DBMS,91
```

```
106,BigData,74
109,DBMS,70
```

Shell commands

```
pig -version
jps
cat data/students.txt
cat data/marks.txt
hdfs dfs -mkdir -p /bdt/bucket8
hdfs dfs -put -f data/students.txt data/marks.txt /bdt/bucket8/
hdfs dfs -ls /bdt/bucket8
```

Step 2 – Pig Latin script

bucket8/pig/03_sort.pig

```
-- Bucket 8 | Sub-Experiment 3 : Sort data with ORDER BY and LIMIT
students = LOAD '/bdt/bucket8/students.txt' USING PigStorage(',') AS (id:int, name:chararray,
dept:chararray);
marks    = LOAD '/bdt/bucket8/marks.txt'    USING PigStorage(',') AS (id:int, subject:chararray,
marks:int);
joined = JOIN students BY id, marks BY id;
res    = FOREACH joined GENERATE students::name AS name, marks::subject AS subject, marks::marks AS
marks;

-- ascending order of marks (lowest first)
asc_marks  = ORDER res BY marks ASC;
DUMP asc_marks;

-- descending order of marks (highest first)
desc_marks = ORDER res BY marks DESC;
DUMP desc_marks;

-- top 3 scores: ORDER BY + LIMIT
top3 = LIMIT desc_marks 3;
DUMP top3;

-- students sorted by name (A to Z)
by_name = ORDER students BY name ASC;
DUMP by_name;
```

Command

```
pig -x mapreduce -4 ~/bdt/pig/log4j.properties -f pig/03_sort.pig
```

F. Execution Screenshots

Screenshots of the Ubuntu VM terminal (`ubuntu@bigdata`) while `run_bucket8.sh` was executing. Only empty margins were trimmed.

```
BUCKET 8 : Apache Pig - Join, Group, Sort and Filter
Part 1 : Setup: environment check, sample datasets and upload to HDFS

$ pig -version
Apache Pig version 0.18.0 (r1926484)
compiled Jun 16 2025, 10:38:21
$ jps
1696 NameNode
52177 Jps
3377 JobHistoryServer
2483 ResourceManager
1916 DataNode
2157 SecondaryNameNode
2687 NodeManager
$ cat data/students.txt
101,Aarav,Computer
102,Diya,IT
103,Kabir,Computer
104,Meera,Mechanical
105,Rohan,IT
106,Sneha,Computer
107,Arjun,Mechanical
108,Isha,IT
$ cat data/marks.txt
101,DBMS,78
101,BigData,85
102,DBMS,45
102,BigData,62
103,DBMS,91
103,BigData,88
104,DBMS,38
104,BigData,55
105,DBMS,67
105,BigData,49
106,DBMS,82
106,BigData,74
107,DBMS,59
107,BigData,41
103,DBMS,91
106,BigData,74
109,DBMS,70
$ hdfs dfs -mkdir -p /bdt/bucket8
$ hdfs dfs -put -f data/students.txt data/marks.txt /bdt/bucket8/
$ hdfs dfs -ls /bdt/bucket8
Found 2 items
-rw-r--r--   1 ubuntu supergroup        228 2026-10-25 23:36 /bdt/bucket8/marks.txt
-rw-r--r--   1 ubuntu supergroup        136 2026-10-25 23:36 /bdt/bucket8/students.txt
```

Figure 3.1: Environment check, datasets and upload to HDFS

`pig -version` reports Apache Pig 0.18.0 and `jps` lists all Hadoop daemons including the JobHistoryServer. Both datasets are printed and copied to `/bdt/bucket8/` (marks.txt 228 B, students.txt 136 B).

```
BUCKET 8 : Apache Pig – Join, Group, Sort and Filter
Part 4 : Sub-Exp 3 : Sort Data – ORDER BY ascending / descending and LIMIT

$ cat pig/03_sort.pig
-- Bucket 8 | Sub-Experiment 3 : Sort data with ORDER BY and LIMIT
students = LOAD '/bdt/bucket8/students.txt' USING PigStorage(',') AS (id:int, name:chararray, dept:chararray);
marks    = LOAD '/bdt/bucket8/marks.txt'    USING PigStorage(',') AS (id:int, subject:chararray, marks:int);
joined = JOIN students BY id, marks BY id;
res    = FOREACH joined GENERATE students::name AS name, marks::subject AS subject, marks::marks AS marks;

-- ascending order of marks (lowest first)
asc_marks  = ORDER res BY marks ASC;
DUMP asc_marks;

-- descending order of marks (highest first)
desc_marks = ORDER res BY marks DESC;
DUMP desc_marks;

-- top 3 scores: ORDER BY + LIMIT
top3 = LIMIT desc_marks 3;
DUMP top3;

-- students sorted by name (A to Z)
by_name = ORDER students BY name ASC;
DUMP by_name;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/03_sort.pig
2026-10-25 23:39:06,660 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 23:39:06,661 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 23:39:06,661 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 23:39:06,689 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
23:39:06 [INFO] Main – Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
23:39:06 [INFO] Main – Logging error messages to: /home/ubuntu/bdt/bucket8/pig_1792951746687.log
23:39:06 [INFO] HExecutionEngine – Connecting to hadoop file system at: hdfs://localhost:9000
23:39:07 [WARN] PigServer – ATS is disabled since yarn.timeline-service.enabled set to false
23:39:08 [INFO] MapReduceLauncher – 1 map-reduce job(s) waiting for submission.
23:39:08 [WARN] JobResourceUploader – No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:39:09 [INFO] MapReduceLauncher – HadoopJobId: job_1792936583670_0093
23:39:09 [INFO] MapReduceLauncher – Processing aliases joined,marks,res,students
23:39:09 [INFO] MapReduceLauncher – detailed locations: M: marks[3,11],marks[-1,-1],joined[4,9],students[2,11],students[-1,-1],joined[4,9] C:  R: res[5,9]
23:39:09 [INFO] MapReduceLauncher – 0% complete
23:39:09 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0093]
23:39:16 [INFO] MapReduceLauncher – 16% complete
23:39:16 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0093]
23:39:21 [INFO] MapReduceLauncher – 33% complete
23:39:21 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0093]
23:39:26 [INFO] MapReduceLauncher – 1 map-reduce job(s) waiting for submission.
23:39:26 [WARN] JobResourceUploader – No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:39:26 [INFO] MapReduceLauncher – HadoopJobId: job_1792936583670_0094
23:39:26 [INFO] MapReduceLauncher – Processing aliases asc_marks
23:39:26 [INFO] MapReduceLauncher – detailed locations: M: asc_marks[8,13] C:  R:
23:39:36 [INFO] MapReduceLauncher – 50% complete
23:39:36 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0094]
23:39:39 [INFO] MapReduceLauncher – 66% complete
23:39:39 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0094]
23:39:44 [INFO] MapReduceLauncher – 1 map-reduce job(s) waiting for submission.
23:39:44 [WARN] JobResourceUploader – No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:39:44 [INFO] MapReduceLauncher – HadoopJobId: job_1792936583670_0095
23:39:44 [INFO] MapReduceLauncher – Processing aliases asc_marks
23:39:44 [INFO] MapReduceLauncher – detailed locations: M: asc_marks[8,13] C:  R:
23:39:54 [INFO] MapReduceLauncher – 83% complete
23:39:54 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0095]
23:39:59 [INFO] MapReduceLauncher – Running jobs are [job_1792936583670_0095]
23:40:00 [INFO] MapReduceLauncher – 100% complete
23:40:00 [INFO] SimplePigStats – Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 23:39:07     2026-10-25 23:40:00     HASH_JOIN,ORDER_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0093  2       1       2       1       2       2       1       1       1       1       joined,marks,res,students       HASH_JOIN
job_1792936583670_0094  1       1       1       1       1       1       1       1       1       1       asc_marks       SAMPLER
```

Figure 3.2: Sort script and the ascending-sort jobs

The script joins students with marks, projects (name, subject, marks) and sorts. The ascending sort runs three chained jobs: `_0093` HASH_JOIN, `_0094` SAMPLER, `_0095` ORDER_BY.

```
Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0093  2       1       2       1       2       2       1       1       1       1       joined,marks,res,students       HASH_JOIN
job_1792936583670_0094  1       1       1       1       1       1       1       1       1       1       asc_marks       SAMPLER
job_1792936583670_0095  1       1       1       1       1       1       2       2       2       2       asc_marks       ORDER_BY        hdfs://localhost:9000/tmp/temp-482489939/tmp-1219702401,

Input(s):
Successfully read 8 records from: "/bdt/bucket8/students.txt"
Successfully read 17 records from: "/bdt/bucket8/marks.txt"

Output(s):
Successfully stored 16 records (320 bytes) in: "hdfs://localhost:9000/tmp/temp-482489939/tmp-1219702401"

Counters:
Total records written : 16
Total bytes written : 320
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0093  ->      job_1792936583670_0094,
job_1792936583670_0094  ->      job_1792936583670_0095,
job_1792936583670_0095


23:40:00 [INFO] MapReduceLauncher - Success!
(Meera,DBMS,38)
(Arjun,BigData,41)
(Diya,DBMS,45)
(Rohan,BigData,49)
(Meera,BigData,55)
(Arjun,DBMS,59)
(Diya,BigData,62)
(Rohan,DBMS,67)
(Sneha,BigData,74)
(Sneha,BigData,74)
(Aarav,DBMS,78)
(Sneha,DBMS,82)
(Aarav,BigData,85)
(Kabir,BigData,88)
(Kabir,DBMS,91)
(Kabir,DBMS,91)
23:40:00 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
23:40:01 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:40:01 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:40:02 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0096
23:40:02 [INFO] MapReduceLauncher - Processing aliases joined,marks,res,students
23:40:02 [INFO] MapReduceLauncher - detailed locations: M: students[2,11],students[-1,-1],joined[4,9],marks[3,11],marks[-1,-1],joined[4,9] C:  R: res[5,9]
23:40:02 [INFO] MapReduceLauncher - 0% complete
23:40:02 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0096]
23:40:14 [INFO] MapReduceLauncher - 16% complete
23:40:14 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0096]
23:40:19 [INFO] MapReduceLauncher - 33% complete
23:40:19 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0096]
23:40:24 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:40:24 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:40:25 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0097
23:40:25 [INFO] MapReduceLauncher - Processing aliases desc_marks
23:40:25 [INFO] MapReduceLauncher - detailed locations: M: desc_marks[12,13] C:  R:
23:40:32 [INFO] MapReduceLauncher - 50% complete
23:40:32 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0097]
23:40:37 [INFO] MapReduceLauncher - 66% complete
23:40:37 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0097]
23:40:41 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:40:41 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:40:42 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0098
23:40:42 [INFO] MapReduceLauncher - Processing aliases desc_marks
23:40:42 [INFO] MapReduceLauncher - detailed locations: M: desc_marks[12,13] C:  R:
23:40:52 [INFO] MapReduceLauncher - 83% complete
23:40:52 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0098]
23:40:56 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0098]
```

Figure 3.3: Ascending order of marks

16 records from lowest to highest: (Meera,DBMS,38), (Arjun,BigData,41), (Diya,DBMS,45) ... (Kabir,BigData,88), (Kabir,DBMS,91), (Kabir,DBMS,91). The descending sort jobs start.

```
23:40:57 [INFO] MapReduceLauncher - Success!
(Kabir,DBMS,91)
(Kabir,DBMS,91)
(Kabir,BigData,88)
(Aarav,BigData,85)
(Sneha,DBMS,82)
(Aarav,DBMS,78)
(Sneha,BigData,74)
(Sneha,BigData,74)
(Rohan,DBMS,67)
(Diya,BigData,62)
(Arjun,DBMS,59)
(Meera,BigData,55)
(Rohan,BigData,49)
(Diya,DBMS,45)
(Arjun,BigData,41)
(Meera,DBMS,38)
23:40:57 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
23:40:59 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:40:59 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:41:00 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0099
23:41:00 [INFO] MapReduceLauncher - Processing aliases joined,marks,res,students
23:41:00 [INFO] MapReduceLauncher - detailed locations: M: marks[3,11],marks[-1,-1],joined[4,9],students[2,11],students[-1,-1],joined[4,9] C:  R: res[5,9]
23:41:00 [INFO] MapReduceLauncher - 0% complete
23:41:00 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0099]
23:41:09 [INFO] MapReduceLauncher - 6% complete
23:41:09 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0099]
23:41:10 [INFO] MapReduceLauncher - 12% complete
23:41:10 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0099]
23:41:15 [INFO] MapReduceLauncher - 25% complete
23:41:15 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0099]
23:41:22 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:41:22 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:41:23 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0100
23:41:23 [INFO] MapReduceLauncher - Processing aliases desc_marks
23:41:23 [INFO] MapReduceLauncher - detailed locations: M: desc_marks[12,13] C:  R:
23:41:30 [INFO] MapReduceLauncher - 37% complete
23:41:30 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0100]
23:41:35 [INFO] MapReduceLauncher - 50% complete
23:41:35 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0100]
23:41:39 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:41:39 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:41:40 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0101
23:41:40 [INFO] MapReduceLauncher - Processing aliases desc_marks
23:41:40 [INFO] MapReduceLauncher - detailed locations: M: desc_marks[12,13] C:  R:
23:41:50 [INFO] MapReduceLauncher - 62% complete
23:41:50 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0101]
23:41:55 [INFO] MapReduceLauncher - 75% complete
23:41:55 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0101]
23:41:57 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:41:57 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:41:58 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0102
23:41:58 [INFO] MapReduceLauncher - Processing aliases desc_marks
23:41:58 [INFO] MapReduceLauncher - detailed locations: M: desc_marks[12,13] C:  R:
23:42:07 [INFO] MapReduceLauncher - 87% complete
23:42:07 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0102]
23:42:13 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0102]
23:42:14 [INFO] MapReduceLauncher - 100% complete
23:42:14 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 23:40:57     2026-10-25 23:42:14     HASH_JOIN,ORDER_BY,LIMIT

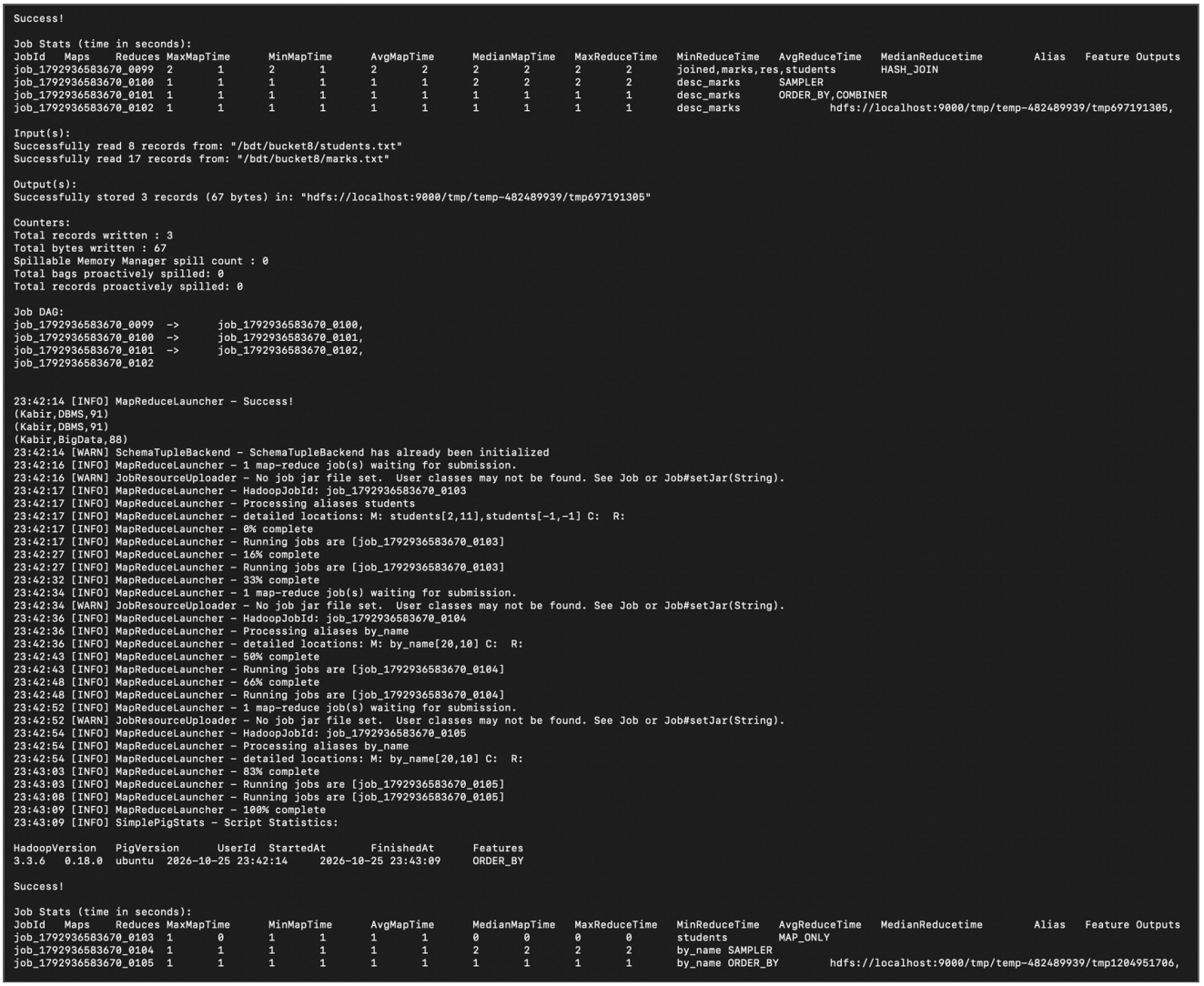
Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0099  2       1       2       1       2       2       2       2       2       2       joined,marks,res,students       HASH_JOIN
job_1792936583670_0100  1       1       1       1       1       1       2       2       2       2       desc_marks      SAMPLER
job_1792936583670_0101  1       1       1       1       1       1       1       1       1       1       desc_marks      ORDER_BY,COMBINER
job_1792936583670_0102  1       1       1       1       1       1       1       1       1       1       desc_marks              hdfs://localhost:9000/tmp/temp-482489939/tmp697191305,

Input(s):
```

Figure 3.4: Descending order of marks

The same 16 records from highest to lowest: (Kabir,DBMS,91) … (Meera,DBMS,38). The top-3 jobs then start.



```
Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0099  2       1       2       1       2       2       2       2       2       2       joined,marks,res,students       HASH_JOIN
job_1792936583670_0100  1       1       1       1       1       1       2       2       2       2       desc_marks      SAMPLER
job_1792936583670_0101  1       1       1       1       1       1       1       1       1       1       desc_marks      ORDER_BY,COMBINER
job_1792936583670_0102  1       1       1       1       1       1       1       1       1       1       desc_marks              hdfs://localhost:9000/tmp/temp-482489939/tmp697191305,

Input(s):
Successfully read 8 records from: "/bdt/bucket8/students.txt"
Successfully read 17 records from: "/bdt/bucket8/marks.txt"

Output(s):
Successfully stored 3 records (67 bytes) in: "hdfs://localhost:9000/tmp/temp-482489939/tmp697191305"

Counters:
Total records written : 3
Total bytes written : 67
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0099  ->      job_1792936583670_0100,
job_1792936583670_0100  ->      job_1792936583670_0101,
job_1792936583670_0101  ->      job_1792936583670_0102,
job_1792936583670_0102


23:42:14 [INFO] MapReduceLauncher - Success!
(Kabir,DBMS,91)
(Kabir,DBMS,91)
(Kabir,BigData,88)
23:42:14 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
23:42:16 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:42:16 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:42:17 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0103
23:42:17 [INFO] MapReduceLauncher - Processing aliases students
23:42:17 [INFO] MapReduceLauncher - detailed locations: M: students[2,11],students[-1,-1] C:  R:
23:42:17 [INFO] MapReduceLauncher - 0% complete
23:42:17 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0103]
23:42:27 [INFO] MapReduceLauncher - 16% complete
23:42:27 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0103]
23:42:32 [INFO] MapReduceLauncher - 33% complete
23:42:34 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:42:34 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:42:36 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0104
23:42:36 [INFO] MapReduceLauncher - Processing aliases by_name
23:42:36 [INFO] MapReduceLauncher - detailed locations: M: by_name[20,10] C:  R:
23:42:43 [INFO] MapReduceLauncher - 50% complete
23:42:43 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0104]
23:42:48 [INFO] MapReduceLauncher - 66% complete
23:42:48 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0104]
23:42:52 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
23:42:52 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
23:42:54 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0105
23:42:54 [INFO] MapReduceLauncher - Processing aliases by_name
23:42:54 [INFO] MapReduceLauncher - detailed locations: M: by_name[20,10] C:  R:
23:43:03 [INFO] MapReduceLauncher - 83% complete
23:43:03 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0105]
23:43:08 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0105]
23:43:09 [INFO] MapReduceLauncher - 100% complete
23:43:09 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 23:42:14     2026-10-25 23:43:09     ORDER_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0103  1       0       1       1       1       1       0       0       0       0       students        MAP_ONLY
job_1792936583670_0104  1       1       1       1       1       1       2       2       2       2       by_name SAMPLER
job_1792936583670_0105  1       1       1       1       1       1       1       1       1       1       by_name ORDER_BY        hdfs://localhost:9000/tmp/temp-482489939/tmp1204951706,
```

Figure 3.5: Top 3 with ORDER BY + LIMIT

Four jobs (HASH_JOIN, SAMPLER, ORDER_BY+COMBINER, LIMIT) store 3 records: (Kabir,DBMS,91), (Kabir,DBMS,91), (Kabir,BigData,88). The sort-by-name jobs start (MAP_ONLY → SAMPLER → ORDER_BY).

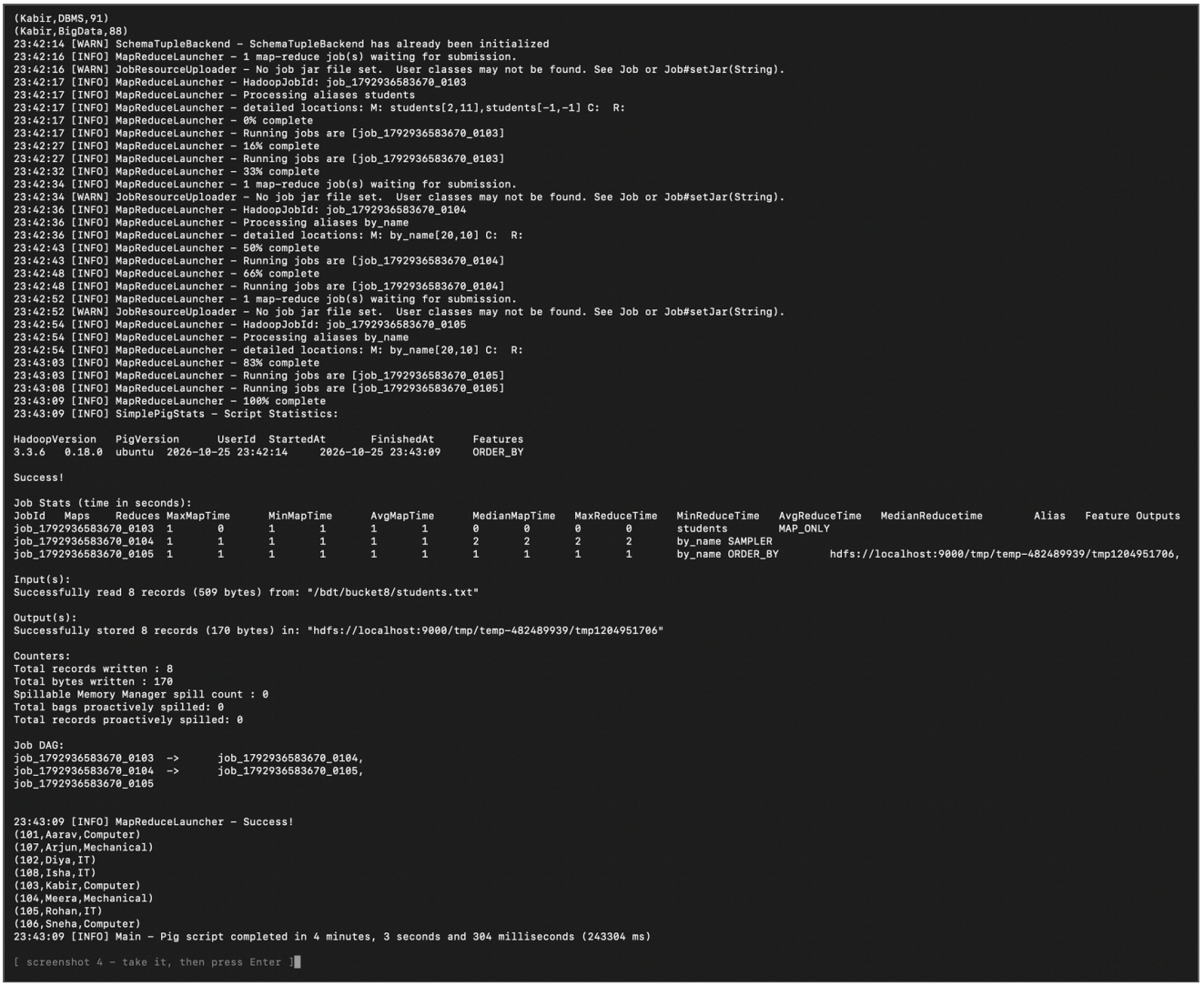


Figure 3.6: Students sorted by name (A to Z)

8 students in alphabetical order: Aarav, Arjun, Diya, Isha, Kabir, Meera, Rohan, Sneha. The whole script completes in 4 min 3 s.

G. Output and Results

Rank	Ascending (lowest first)	Descending (highest first)
1	Meera, DBMS, 38	Kabir, DBMS, 91
2	Arjun, BigData, 41	Kabir, DBMS, 91
3	Diya, DBMS, 45	Kabir, BigData, 88
4	Rohan, BigData, 49	Aarav, BigData, 85
…	…	…
16	Kabir, DBMS, 91	Meera, DBMS, 38

Query	Result
Top 3 (DESC + LIMIT 3)	(Kabir,DBMS,91) (Kabir,DBMS,91) (Kabir,BigData,88)
ORDER students BY name ASC	Aarav, Arjun, Diya, Isha, Kabir, Meera, Rohan, Sneha

The descending result is exactly the ascending result reversed, which shows the effect of `ASC` vs `DESC`. The top-3 list contains Kabir's DBMS score twice because `marks.txt` contains that row twice.

H. Result

> The objective was achieved. `ORDER BY marks ASC` and `DESC` sorted all 16 joined records in opposite orders, `LIMIT 3` after a descending sort returned the top three scores, and `ORDER BY name` sorted the students alphabetically.

I. Observations and Troubleshooting

Observations

- Every ORDER BY needed a sampler job plus a sort job, so sorting is the most expensive operation here (the script took about 4 minutes).
- Records with equal marks (e.g. Sneha 74 twice) appear next to each other in no guaranteed order.
- Sorting by name is case-sensitive: upper-case letters sort before lower-case ones.

Setup notes

- Apache Pig 0.18.0 (Hadoop 3 build) with `PIG_HOME`, `PATH` and `PIG_CLASSPATH` set in `~/.bashrc`; the MapReduce JobHistory server must be running because Pig requests job statistics from it.
- The warnings *No job jar file set*, *ATS is disabled* and *SchemaTupleBackend has already been initialized* are harmless in this pseudo-distributed setup.
- The script ran once without errors; no troubleshooting was required.